Stealth Attacks on Swarms: Adversary Modeling from a Zero-Sum Game Perspective

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1 Problem Statement

We study a *rational, stealthy attacker* acting against a swarm under partial observability with inter-agent communication. We refer to the swarm as "team blue" (the n nominal agents indexed by $\mathcal{I} = \{1, \ldots, n\}$), and to the attacker as "team red" (a rational opponent that can influence a limited number of blue agents). In the absence of an attacker, the blue-team mission is *collaborative*: agents may pursue different local goals or subtasks, but their coordinated behavior and task fusion define the overall mission objective. We model this baseline as a decentralized partially observable Markov decision process (Dec-POMDP), or equivalently as a common-payoff partially observable stochastic game (POSG), because all blue agents ultimately optimize a shared mission return (see, e.g., Hansen et al. [10]).

To reason about *rational* worst-case interference, we adopt the POSG lens and a standard game-theoretic viewpoint in which red and blue have opposing objectives [1, 19]. In this project, the POSG formalism is used primarily to express this threat model (information, intervention channels, and stealth constraints), rather than to fully solve for equilibrium policies. We now define the corresponding Dec-POMDP/POSG tuple:

$$\mathcal{G} = \langle \mathcal{I}, \mathcal{S}, \{\mathcal{A}_i\}_{i\in\mathcal{I}}, \{\mathcal{O}_i\}_{i\in\mathcal{I}}, P, O, R, \gamma, \rho^0 \rangle. \tag{1}$$

The tuple in (1) lists the standard ingredients of a Dec-POMDP, which is equivalently an POSG:

- **Agents $\mathcal{I} = \{1, \ldots, n\}$.** The decision makers (robots). In POSG language, these are the *players*.
- **State space $\mathcal{S}$.** The hidden "ground truth" of the environment (e.g., robot poses, targets, obstacles). In a POSG/Dec-POMDP this is the global state that drives the dynamics.
- **Action spaces $\{\mathcal{A}_i\}_{i\in\mathcal{I}}$.** What each agent can do at a time step (its available controls). The joint action is $\mathbf{a}_t \in \mathcal{A} \triangleq \prod_{i\in\mathcal{I}} \mathcal{A}_i$.
- **Observation spaces $\{\mathcal{O}_i\}_{i\in\mathcal{I}}$.** What each agent can measure (its sensor readings). The joint observation is $\mathbf{o}_t \in \mathcal{O} \triangleq \prod_{i\in\mathcal{I}} \mathcal{O}_i$.
- **Transition model $P(s' \mid s, \mathbf{a})$.** How the world changes when the swarm takes a joint action (the system dynamics). This is the same object in Dec-POMDPs and POSGs.
- **Observation model $O(\mathbf{o} \mid s', \mathbf{a})$.** How the (hidden) next state produces the agents' observations (the sensing/partial-observability model). Dec-POMDPs often assume a factored form across agents; POSGs may be written either factored or joint.
- **Reward $R(s, \mathbf{a}, s')$.** What the team is trying to maximize. In a cooperative Dec-POMDP, the reward is shared by definition; in POSG terms, this is the identical-payoff (common-reward) special case.
- **Discount factor $\gamma \in (0, 1)$.** How much future reward matters relative to immediate reward.
- **Initial distribution $\rho^0 \in \Delta(\mathcal{S})$.** Uncertainty about the starting state ($s_0 \sim \rho^0$).

Communication augments the Dec-POMDP/POSG loop by adding an explicit message-passing layer on top of sensing and control: at time t, each agent i broadcasts a message $m_{i,t} \in \mathcal{M}$ (e.g., a local estimate, intent, or learned embedding), and then forms an incoming summary $\bar{m}_{i,t} = \mathsf{Agg}_i(\{m_{j,t} : j \in \mathcal{N}_i(t)\})$ from the messages received from its current neighbors $\mathcal{N}_i(t)$. The neighbor sets are determined by a time-varying communication graph $G_t = (\mathcal{I}, E_t)$, so which agents can exchange information may change over time due to motion, range limits, packet loss, or link failures. Relative to the baseline

state–observation–action structure in Fig. 1, messages provide an additional information channel that can improve coordination under partial observability, but also creates a structured attack surface. In practice, an external attacker may not know the instantaneous edge set E_t (and may only have partial access to the swarm's internal states), which makes direct targeting based on global graph centrality or rank unreliable without first learning or probing the network. This motivates black-box and online influence-estimation attacker models that operate under incomplete knowledge of the communication structure [23, 22].

Each agent follows a decentralized policy: at time t, agent i selects $a_{i,t}$ using only its own local history and the messages it receives,

$$a_{i,t} \sim \pi_i(\tau_{i,t}, \bar{m}_{i,t}), \qquad \tau_{i,t} = \{o_{i,0}, m_{i,0}, a_{i,0}, \ldots, o_{i,t}\}.$$

This emphasizes the key Dec-POMDP constraint: there is no centralized controller with access to the full state s_t or the full joint observation $\mathbf{o}_t$.

In contrast, we treat the attacker as *centralized* in the sense that it can coordinate its interventions across multiple compromised agents and time steps toward a single objective (mission degradation), even if it does not know the full state or network topology. This asymmetry, a decentralized swarm facing a coordinated adversary, captures realistic compromise scenarios in which malware, a human operator, or a single planning module can synchronize the behavior of several infected robots, while the healthy robots remain distributed and must act from local information. It is also an *imbalanced* game: the blue team contains all n agents, whereas the red team can influence at most k agents with $k \ll n$ [21].

We model a *stealthy, centralized* attacker that compromises an unknown subset $\mathcal{C} \subseteq \mathcal{I}$ of at most k agents and *changes how those agents behave* at execution time (e.g., by manipulating their local policy inputs, internal state, or action selection), with the goal of degrading the team return. Importantly, the communication mechanism itself is not assumed to be tampered with: uncompromised agents receive whatever messages are broadcast, but the compromised agents may generate *plausible* messages and actions that steer the swarm toward poor outcomes.

We capture "stealth" by constraining how far the compromised behavior can deviate from nominal, so that the agent itself (and simple consistency checks by neighbors) cannot easily flag the compromise. Concretely, let π_i denote the nominal policy and $\tilde{\pi}_i$ the behavior executed by a compromised agent; an attacker strategy ϕ selects $\mathcal{C}$ and induces $\tilde{\pi}_i$ for $i \in \mathcal{C}$ subject to a deviation budget, e.g.,

$$d(\tilde{\pi}_i(\cdot \mid \tau_{i,t}, \bar{m}_{i,t}),\, \pi_i(\cdot \mid \tau_{i,t}, \bar{m}_{i,t})) \le \varepsilon, \qquad \sum_t \mathbb{I}[\tilde{\pi}_i(\cdot \mid \tau_{i,t}, \bar{m}_{i,t}) \neq \pi_i(\cdot \mid \tau_{i,t}, \bar{m}_{i,t})] \le T_{\text{int}},$$

where $d(\cdot,\cdot)$ measures how "noticeable" the policy/action change is (e.g., a norm bound on action perturbations, a KL bound between action distributions, or a detector score), ε limits per-step deviation, and T_{int} limits how often team red intervenes (an *intervention budget*). Team blue's goals are (i) mission-level robustness (small performance loss) and (ii) detectability and localization of compromised agents, while team red is interested in the complementary question: what is the minimum compromise size k needed to cause a mission degradation of at least θ:

$$\begin{aligned}
&\textbf{(Team blue objective)} \quad J(\pi,\phi) \triangleq \mathbb{E}\Big[\sum_{t\ge 0} \gamma^t r_t\Big], \\
&\qquad\qquad\qquad\qquad \pi^\star \in \arg\max_{\pi} \min_{\phi\in\Phi_k} J(\pi,\phi), \\
&\textbf{(Team red objective)} \quad J_{\text{red}}(\pi,\phi) \triangleq -J(\pi,\phi), \\
&\qquad\qquad\qquad\qquad \phi^\star \in \arg\min_{\phi\in\Phi_k} J(\pi,\phi),
\end{aligned} \tag{2}$$

where $\phi \in \Phi_k$ denotes a feasible team-red strategy that compromises at most k agents ($|\mathcal{C}| \le k$) and respects the stealth budgets $(\varepsilon, T_{\text{int}})$.

$$\begin{aligned}
\Delta J(k) &\triangleq J(\pi;\text{clean}) - \min_{\phi\in\Phi_k} J(\pi,\phi), \\
k^\star(\theta) &\triangleq \min\Big\{k:\ \Delta J(k) \ge \theta\Big\}.
\end{aligned} \tag{3}$$

2 Related Work

Our setting is a collaborative swarm operating under partial observability and inter-agent communication, where performance is ultimately defined by a shared mission return. Decentralized POMDPs

(Dec-POMDPs) are a standard formalism for such collaborative teams with decentralized information, and classical complexity results clarify why practical approaches typically rely on approximations and structural assumptions [2]. At a broader level, partially observable stochastic games (POSGs) capture multi-agent sequential decision making under partial observability with potentially distinct reward functions [9]. The attack-free baseline we consider aligns with the identical-payoff (common-payoff) subclass of POSGs; related robotics work has used common-payoff POSGs to emphasize how belief evolution, communication, and approximate joint planning jointly shape coordinated behavior [5]. Recent work on one-sided POSGs and related subclasses further bridges fully general POSGs and more tractable pipelines, which is relevant when one side enjoys substantially richer information or coordination [11].

To capture *rational* worst-case interference, it is natural to adopt a dynamic game-theoretic lens in which an adversary (team red) selects interventions to reduce the blue team's mission return. Minimax value concepts and equilibrium viewpoints trace back to classical stochastic games [19] and are developed broadly in dynamic noncooperative game theory [1]. In robotics and learning, pursuit–evasion and Markov-game formulations provide an RL-oriented language for adversarial objectives over trajectories and information [13, 17], while security-game perspectives motivate modeling the attacker as an optimizing agent that allocates limited influence (e.g., a small compromise set) to maximize mission damage [21].

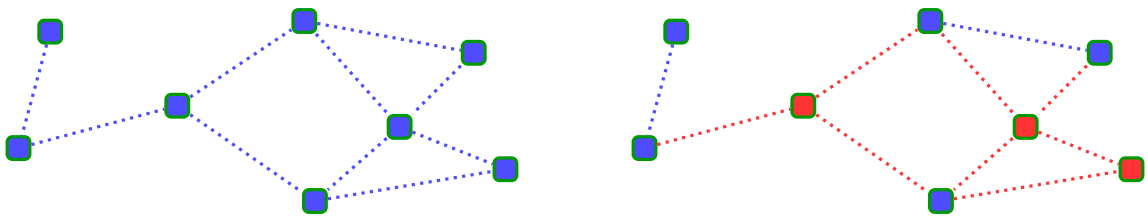
Communication is a primary mechanism for mitigating partial observability in MARL, and recent surveys systematize key design choices in communication-enabled methods [32]. This same message-passing layer is also a high-leverage attack surface. Hijacking results based on adversarial messaging illustrate that black-box manipulation can be feasible even without access to the victim reward or policies [23], and early work on adversarial examples for RL policies provides foundational tools for crafting observation- and input-level perturbations [12]. Beyond generic input corruption, adversarial communication has been studied by explicitly learning malicious messages and proposing reconstruction- and game-theoretic defenses [24]; complementary lines of work pursue certifiable robustness when the communication module and threat set are modeled explicitly [20]. Persistent compromise models include backdoor-style attacks that exploit the observation/communication pathway to induce triggerable failures in cooperative MARL, including attacks tailored to value-decomposition architectures and recent work on stealthy spatiotemporal triggers [3, 25].

Stealth and resource constraints motivate *budgeted* and *sparse* attack models, intervening on few agents and/or few time steps, that can approach the impact of always-on perturbations while being harder to detect [16]. Subset-selection and contribution-aware strategies further connect attacker efficiency to multi-agent credit assignment, including dynamic group-based transferability methods [26] and formalizations of minority influence in cooperative systems [15]. On the defense side, robustification has been explored via adversarial formulations for value factorization (e.g., robust variants of QMIX) [8]. A related but distinct threat model considers *adversarial agent policies* that induce harmful yet "natural" observations through interaction rather than direct sensor/message corruption, emphasizing that interference can arise without explicit channel corruption [6].

Finally, mission degradation under stealth has close analogs in cyber-physical system (CPS) security, including false-data injection and intermittent integrity attacks. Residual-based false data injection illustrates optimization-driven stealth under resource constraints [7], and intermittent integrity attack models formalize detectability under sparse schedules [28]. Related work studies composite detection for stealthy data deception [27] and resilient coordination objectives such as secure multi-dimensional consensus under malicious agents [18]. Across CPS and MARL, "stealth" is often operationalized via explicit perturbation/intermittency budgets together with detector performance (e.g., FAR–TPR curves, AUC, and detection delay), linking constraints such as $\mathbb{P}(\text{detected} \mid \phi) \leq \alpha$ to mission-level impact [7, 28, 27]. In parallel, attribution and interpretability tools can help localize compromised influence, including transfer-driven reward redistribution and counterfactual value-based threat detection in decentralized settings [14, 4]. Finally, to connect these threat models to concrete benchmarks and structured objectives, prior work has (i) encoded mission requirements using temporal logic and derived shaped rewards from automata-based decompositions, and (ii) evaluated coordination and interference effects in canonical pursuit, evasion-style swarm tasks studied with deep MARL [31, 29, 30].

3 Hypothesis

Our hypothesis follows a simple narrative: communication makes cooperative swarms effective under partial observability, but it also creates a structured interface that a rational adversary can exploit. Because blue agents optimize a shared return yet execute decentralized policies, an attacker that can



(a) Blue-team swarm under nominal communication/coordination.

(b) Red-team compromises agents/links to induce stealthy mission failure.

Figure 1: Blue team (defender) and red team (attacker) are modeled as opposing policies in a zero-sum POSG/Markov-game view: the swarm executes a defensive policy π to maximize mission return, while the attacker executes an intervention policy ϕ to minimize it, i.e., $J_{\text{blue}}(\pi, \phi) = -J_{\text{red}}(\pi, \phi)$.

coordinate limited interventions across agents and time can induce mission-level degradation that looks qualitatively different from random faults. This motivates the two-player zero-sum POSG/Markov-game abstraction used here: the blue team executes a defensive policy π to maximize return, while team red chooses an intervention policy $\phi \in \Phi_k(\varepsilon, T_{\text{int}})$ to minimize it.

First, we expect *small compromise sets to have amplified effects* when they target the information flow that drives coordination. Under learned communication, messages can act as shared belief features or intent signals; corrupting a few agents can therefore propagate through aggregation and downstream decision rules, producing minority-to-majority influence effects and disproportionately large return loss [15, 23]. This amplification should be strongest when the compromised agents occupy influential positions in the interaction/communication structure (e.g., bottlenecks or highly relied-upon senders).

Second, we expect a *robustness gap between random failures and adaptive best responses*: the same "amount" of corruption can be far more damaging when it is placed *strategically*. Random message noise or random dropouts spread their effect across agents and time; by contrast, an adaptive attacker can spend its limited budget to intervene only when (and where) the swarm is most sensitive, e.g., near decision boundaries, during critical coordination phases, or on agents whose messages strongly influence others. Under the same budgets $(k, \varepsilon, T_{\text{int}})$, this targeted placement can therefore produce substantially larger return loss than naive corruption, while still remaining stealthy by acting sparsely in time and keeping each intervention small [16, 26, 28, 7].

Finally, our goal is not to prescribe defenses in this section but to motivate a concrete *threat-modeling* agenda. We focus on how a rational, budgeted attacker can select a small compromise set and coordinate interventions over time to induce worst-case mission degradation, yielding an imbalanced game between a decentralized blue team and a centralized red team. This framing sets up the parameters and constraints of interest—$(k, \varepsilon, T_{\text{int}}, \alpha)$ and the attacker information structure—and provides a consistent lens for evaluating "worst-case plausible" interventions as approximate best responses.

Discussion and research questions. We use the POSG/Markov-game view primarily as a *threat-modeling lens*: it specifies what team red can observe, which channels it can manipulate, and which constraints make an intervention stealthy, notably compromise size k, per-step distortion budget ε, an intervention budget T_{int}, and an allowable detectability level α. Rather than solving the full game, the zero-sum framing provides a principled way to define and evaluate
emphplausible worst-case interventions (approximate best responses) and to quantify robustness gaps relative to random faults.

This perspective leads to the following concrete research questions:
We view these questions as a menu of threat-modeling directions; in this work we do not attempt to answer all of them simultaneously, but use them to motivate the modeling choices and evaluation axes that follow.

1. **Minimum red-team footprint for a target mission loss.** What is the smallest compromise size $k^\star(\theta)$ required to achieve $\Delta J(k) \geq \theta$, and how does $k^\star$ scale with swarm size n and communication connectivity?

2. **Stealthy worst-case attacks.** Under per-step distortion budget ε, intervention budget T_{int}, and detectability level α, what is the achievable tradeoff between mission degradation and stealth (e.g., the Pareto frontier between ΔJ and detection probability/delay)?

3. **Which attack surfaces dominate in practice?** When red can influence at most k agents, which intervention channels (messages vs. observations vs. actions) produce the largest worst-case degradation under the same budgets, and how does this depend on the attacker information (e.g., knowledge of policies or the communication graph)?

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